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Java SE 27 & JDK 27

Class PipedReader

java.lang.Object
    java.io.Reader
        java.io.PipedReader

All Implemented Interfaces:

Closeable, AutoCloseable, Readable

```
public class PipedReader
extends Reader
```

Piped character-input streams.

Since:

1.1

Field Summary

Fields declared in class Reader
lock

Constructor Summary

Constructors

Constructor	Description
PipedReader()	Creates a PipedReader so that it is not yet connected.
PipedReader(int pipeSize)	Creates a PipedReader so that it is not yet connected and uses the specified pipe size for the pipe's buffer.
PipedReader(PipedWriter src)	Creates a PipedReader so that it is connected to the piped writer src.
PipedReader(PipedWriter src, int pipeSize)	Creates a PipedReader so that it is connected to the piped writer src and uses the specified pipe size for the pipe's buffer.

Method Summary

All Methods	Instance Methods	

Modifier and Type	Method	Description
void	close()	Closes this piped stream and releases any system resources associated with the stream.
void	connect (PipedWriter src)	Causes this piped reader to be connected to the piped writer src.
int	read()	Reads the next character of data from this piped stream.
int	read(char[] cbuf, int off, int len)	Reads characters into a portion of an array.
boolean	ready()	Tell whether this stream is ready to be read.

Methods declared in class Reader

mark, markSupported, nullReader, of, read, read, readAllAsString, readAllLines, reset, skip, transferTo

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Details

PipedReader

```
public PipedReader(PipedWriter src)
            throws IOException
```

Creates a PipedReader so that it is connected to the piped writer src. Data written to src will then be available as input from this stream.

Parameters:

src - the stream to connect to.

Throws:

IOException - if an I/O error occurs.

PipedReader

```
public PipedReader(PipedWriter src,
                   int pipeSize)
            throws IOException
```

Creates a PipedReader so that it is connected to the piped writer src and uses the specified pipe size for the pipe's buffer. Data written to src will then be available as input from this stream.

Parameters:

src - the stream to connect to.

pipeSize - the size of the pipe's buffer.

Throws:

IOException - if an I/O error occurs.

IllegalArgumentException - if pipeSize <= 0.

Since:

1.6

PipedReader

```
public PipedReader()
```

Creates a PipedReader so that it is not yet connected. It must be connected to a PipedWriter before being used.

PipedReader

```
public PipedReader(int pipeSize)
```

Creates a PipedReader so that it is not yet connected and uses the specified pipe size for the pipe's buffer. It must be connected to a PipedWriter before being used.

Parameters:

pipeSize - the size of the pipe's buffer.

Throws:

IllegalArgumentException - if pipeSize <= 0.

Since:

1.6

Method Details

connect

```
public void connect(PipedWriter src)
             throws IOException
```

Causes this piped reader to be connected to the piped writer `src`. If this object is already connected to some other piped writer, an `IOException` is thrown.

If `src` is an unconnected piped writer and `snk` is an unconnected piped reader, they may be connected by either the call:

```
    snk.connect(src)
```

or the call:

```
    src.connect(snk)
```

The two calls have the same effect.

Parameters:

`src` - The piped writer to connect to.

Throws:

`IOException` - if an I/O error occurs.

read

```
public int read()
         throws IOException
```

Reads the next character of data from this piped stream. If no character is available because the end of the stream has been reached, the value `-1` is returned. This method blocks until input data is available, the end of the stream is detected, or an exception is thrown.

Overrides:

`read` in class `Reader`

Returns:

the next character of data, or `-1` if the end of the stream is reached.

Throws:

`IOException` - if the pipe is `broken`, `unconnected`, closed, or an I/O error occurs.

read

```
public int read(char[] cbuf,
                int off,
                int len)
         throws IOException
```

Reads characters into a portion of an array. This method will block until some input is available, an I/O error occurs, or the end of the stream is reached.

If `len` is zero, then no characters are read and `0` is returned; otherwise, there is an attempt to read at least one character. If no character is available because the stream is at its end, the value `-1` is returned; otherwise, at least one character is read and stored into `cbuf`.

Fewer than `len` characters will be read if `len` exceeds the pipe's buffer size.

Specified by:

`read` in class `Reader`

Parameters:

`cbuf` - Destination buffer

`off` - Offset at which to start storing characters

`len` - Maximum number of characters to read

Returns:

The number of characters read, or -1 if the end of the stream has been reached

Throws:

`IndexOutOfBoundsException` - If `off` is negative, or `len` is negative, or `len` is greater than `cbuf.length - off`

`IOException` - if the pipe is `broken`, `unconnected`, closed, or an I/O error occurs.

ready

```
public boolean ready()
              throws IOException
```

Tell whether this stream is ready to be read. A piped character stream is ready if the circular buffer is not empty.

Overrides:

`ready` in class `Reader`

Returns:

True if the next read() is guaranteed not to block for input, false otherwise. Note that returning false does not guarantee that the next read will block.

Throws:

`IOException` - if the pipe is `broken`, `unconnected`, or closed.

close

```
public void close()
           throws IOException
```

Closes this piped stream and releases any system resources associated with the stream.

Specified by:

`close` in interface `AutoCloseable`

Specified by:

`close` in interface `Closeable`

Specified by:

`close` in class `Reader`

Throws:

`IOException` - if an I/O error occurs.

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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